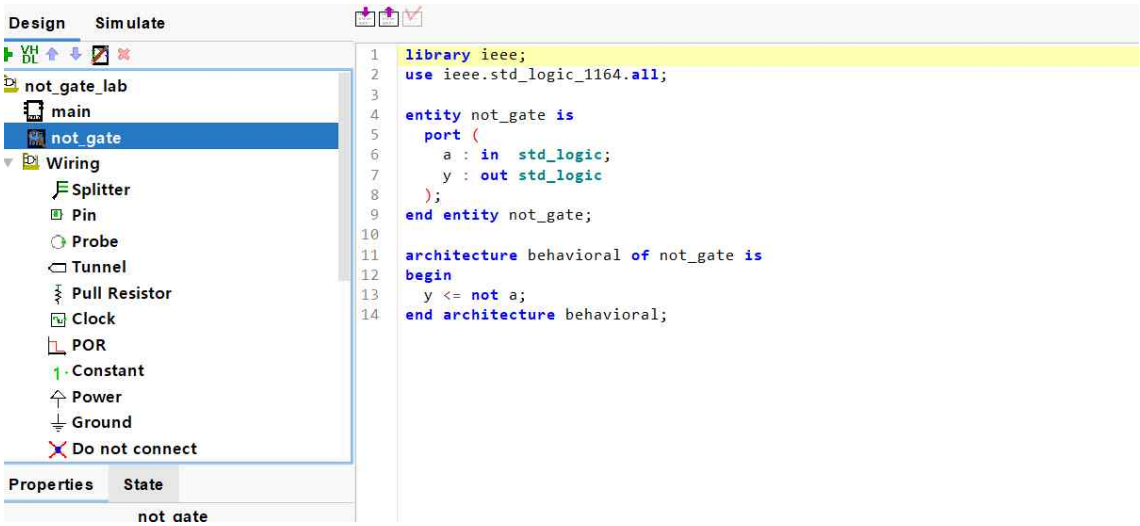
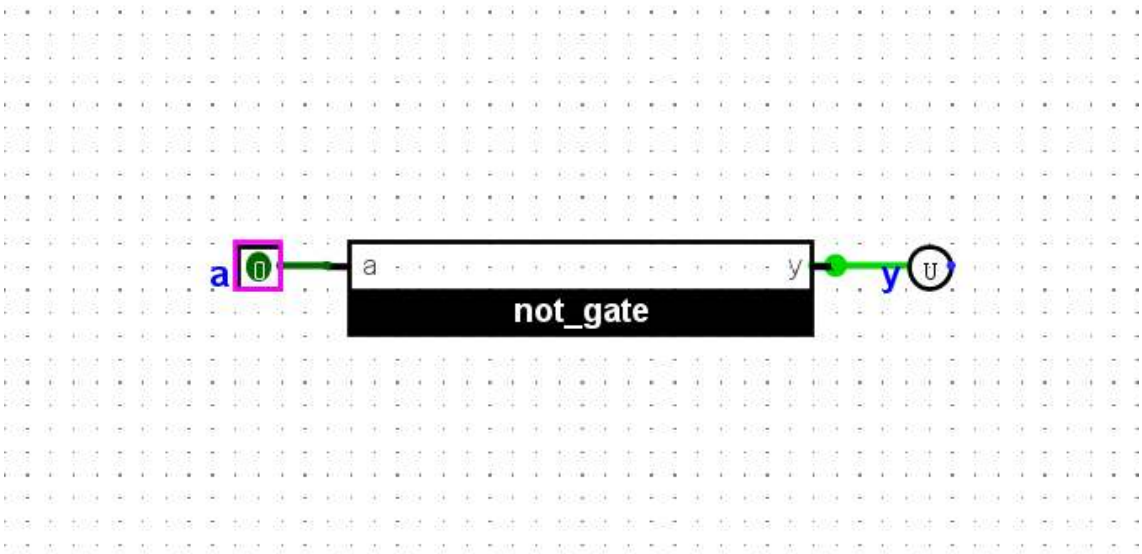
VHDL Assingment

2023048259 정수빈
Robotic Engineering

1. VHDL code



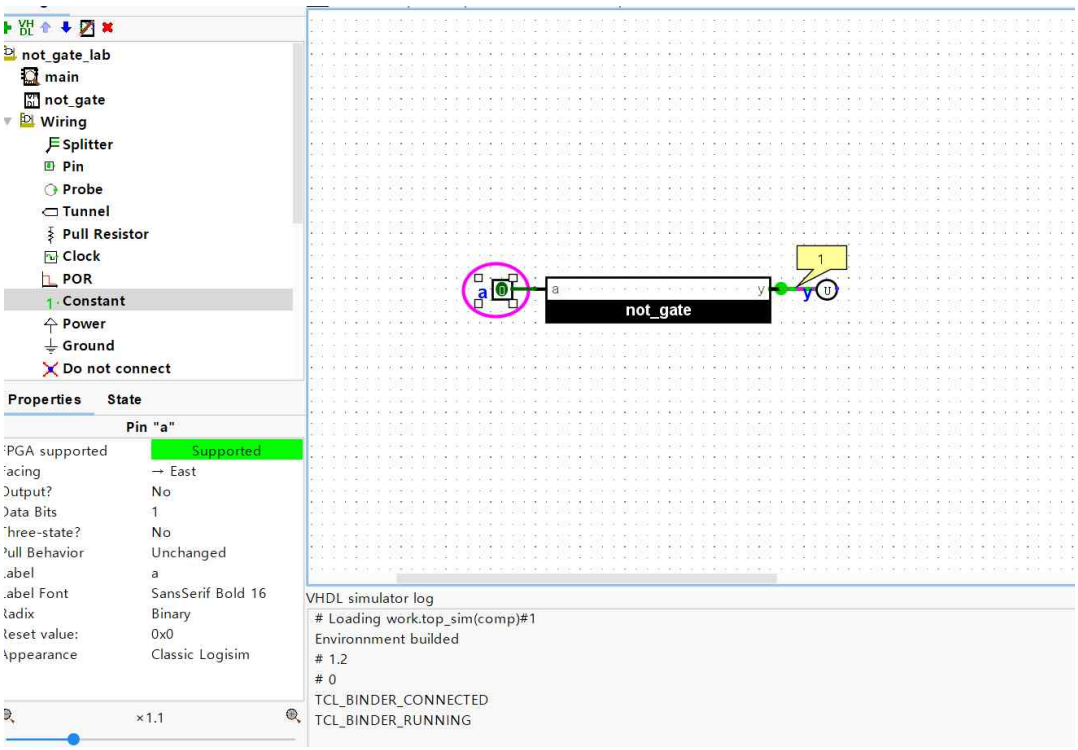
2. circuit



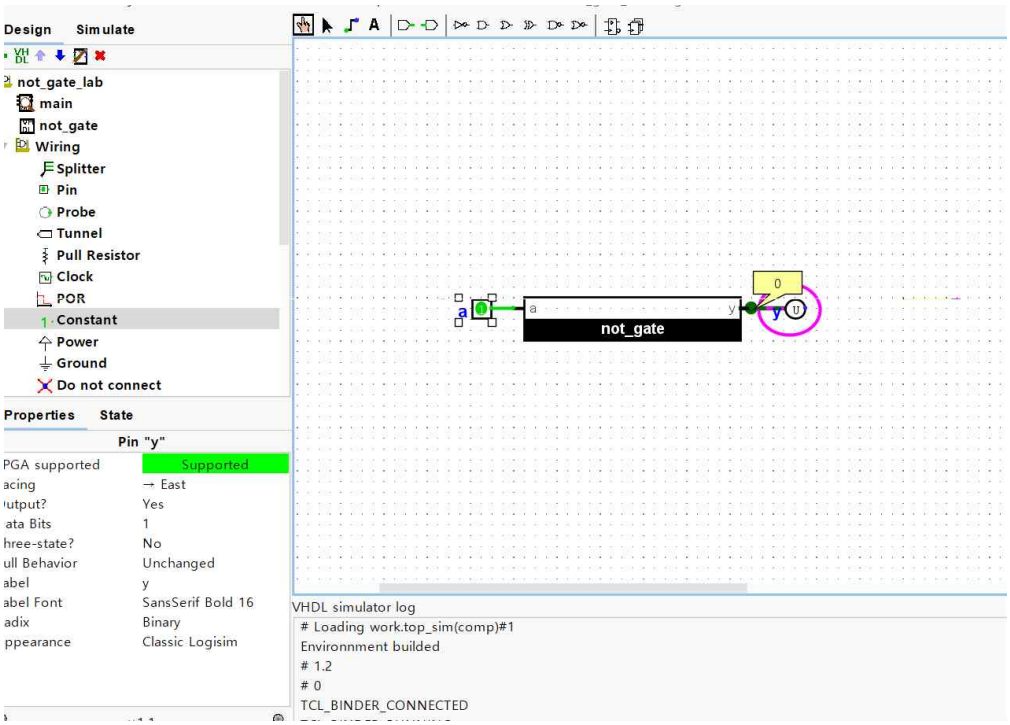
The VHDL not_gate component was placed, and input pin a and output pin y were wired to it.

3. Simulation screenshots

(1) a=0 y=1



(2) a=1 y=0



4. Result

When the input a was set to 0, the output y was 1. When a was set to 1, y was 0. These results match the truth table of a NOT gate, so the VHDL code y <= not a; works as intended.

Input a	Expected output y	Simulated result
0	1	match
1	0	match